A Critical Review of AI-Based Cooperative Selling Optimization for Farmer Producer Organizations: Towards Smarter Post-Harvest Decisions

Arijit Singh1, Tushar Garg2, Hitesh Mehandiratta3, and Namya Gautam4
1,2,4 Department of Computer Engineering, Thapar Institute of Engineering and Technology, Patiala, India
3 Department of Electronics and Communication Engineering, Thapar Institute of Engineering and Technology, Patiala, India
Roll Nos.: 1023039161, 1023036742, 1023152723, 1023038484
Under the Mentorship of Dr. Harkiran Kaur, Assistant Professor, CSED
CPG No. 185 | BE Third Year
Corresponding Author: Arijit Singh (asingh28_be23@thapar.edu)

Abstract. Post-harvest selling inefficiency remains a leading cause of revenue loss for Indian Farmer Producer Organizations (FPOs), resulting in an estimated ₹92,000 crore in annual losses. Recent advancements in graph neural networks, IoT-based monitoring, and cooperative optimization have independently demonstrated strong capabilities in spatiotemporal forecasting, real-time anomaly detection, and multi-agent coordination. However, these technologies have not been integrated into a unified platform that jointly optimises where, when, and how much grain to sell. This paper provides a critical review of the current state-of-the-art techniques across six research streams: agricultural price forecasting, spatiotemporal graph neural networks, IoT grain storage monitoring, post-harvest decision support, cooperative game theory for supply chains, and multilingual farmer advisory systems. The core contribution is a synthesised analysis that identifies the gap between isolated component-level research and the need for an integrated, deployable cooperative selling platform. It highlights current limitations including the absence of spatially aware forecasting for Indian mandi networks, the disconnection between storage monitoring and selling decisions, and the lack of operational scheduling algorithms for FPO coordination. Future research directions are suggested, focusing on developing A3T-GCN-based mandi price models, IoT-constrained selling optimisers, and region-aware multilingual advisory interfaces.

Keywords: Cooperative Selling Optimisation, Graph Neural Networks, A3T-GCN, IoT Grain Monitoring, Farmer Producer Organizations, Post-Harvest Loss, Explainable AI, Agricultural Chatbot

1 Introduction

Post-harvest selling inefficiency remains one of the most significant contributors to revenue loss for smallholder farmers in India. A recent ICRIER study surveying 1,200 farmers across Punjab, Madhya Pradesh, and Bihar found that soybean exhibited total post-harvest losses of 15.34%, and critically, each kilometre increase in distance from the nearest mandi increases paddy loss by 1.4% [1]. These losses, estimated at over ₹1.53 trillion annually at the national level, are driven by fragmented mandi selection, uncoordinated selling behaviour, and inadequate storage

monitoring. Therefore, the development of reliable and integrated decision-support platforms for Farmer Producer Organizations (FPOs) is essential for advancing agricultural market efficiency, reducing avoidable losses, and strengthening the collective bargaining power that FPOs were designed to provide. With growing recognition of FPOs as institutional vehicles for small-farmer empowerment, the study of cooperative selling behaviour has emerged as a critical research area due to its direct impact on farmer income, market stability, and post-harvest waste reduction.

Research in this domain has evolved significantly over the last five years. It has progressed from early statistical time-series models that relied on ARIMA and SVR baselines to sophisticated deep learning architectures capable of capturing complex non-linear price dynamics. Recent benchmarking studies on Indian commodity data have demonstrated that deep learning models, particularly LSTM and GRU, consistently outperform traditional methods by 30–76% on RMSE metrics across volatile agricultural commodities [2]. Concurrently, the emergence of graph neural networks (GNNs) for spatiotemporal forecasting has opened a new paradigm for modelling spatial dependencies across networked systems. Özden and Bulut (2024) presented the first known application of GNNs to agricultural price forecasting, using StemGNN on Turkish wholesale vegetable prices and outperforming CNN, LSTM, and Random Forest baselines [3]. Min et al. (2025) confirmed that GNN-based models outperform RNN-based models for multivariate agricultural price prediction on South Korean commodity data with weather variables [4]. Meanwhile, Mohammadzadeh et al. (2025) adapted A3T-GCN for air quality forecasting using a cosine similarity-based adjacency matrix, reducing RMSE by 11% over standalone GRU and demonstrating that this adjacency construction method is directly applicable to mandi networks where behavioural price similarity matters more than geographic distance [5].

In addition to that, recent research has emphasised the integration of exogenous variables into price prediction frameworks. Nayak et al. (2024) pioneered TransformerX and NBEATSX architectures incorporating weather data from NASA POWER for Indian vegetable price forecasting, demonstrating that weather-augmented deep learning models are robustly superior to univariate approaches [6]. Hybrid architectures combining temporal convolutional networks with gradient boosting have further improved accuracy, with TCN-XGBoost achieving MAPE of 5.3% on global commodity data [7]. A significant milestone was achieved by Shaheen et al. (2025), who conducted the first known real-time deployment of deep learning for Indian fruit price prediction during the 2025 cherry season, achieving greater than 92% accuracy across five wholesale markets [8]. More recently, Chaitra and Meena (2026) introduced a VAR-LSTM hybrid incorporating fuel prices as a covariate for Ragi price prediction in Scientific Reports, validating that transport-related variables significantly influence mandi-level pricing [9]. A 2026 MDPI study further demonstrated that integrating logistics costs, demand-supply ratios, and environmental factors yields superior prediction accuracy for Indian food crops [10].

Parallel to the forecasting domain, IoT-based grain storage monitoring has reached technical maturity. ESP32-based systems with DHT22 temperature-humidity sensors and MQ135 gas

sensors have been widely demonstrated for warehouse and silo monitoring [11], [12]. Recent innovations include TinyML-based edge inference that eliminates cloud dependency [13], LoRaWAN-based long-range communication for distributed facilities achieving 94% anomaly detection accuracy [14], and 3D DenseNet-based deep learning frameworks for spatial temperature field prediction in granaries [15]. A March 2026 study from Karnataka introduced a portable Smart Grain Management System designed for continuous 6–12 month monitoring using ESP32 with DHT22 and MQ135 sensors, specifically targeting smallholder farmers [16]. However, in all reviewed systems, sensor data terminates at detection and alerting. No existing IoT monitoring platform connects storage condition data to selling decision logic, leaving a critical gap between monitoring and actionable market intelligence.

From the coordination perspective, cooperative game theory has provided rigorous frameworks for agricultural supply chain cooperation. Bekolli, Guardiola and Meca (2025) introduced a new class of totally balanced cooperative games proving that all agents benefit from cooperation [17]. Modified Shapley value methods with TOPSIS-based corrective factors have been proposed for fair profit allocation in agricultural supply chains [18], and Stackelberg game models have demonstrated that integrated cooperation yields the highest profits but collapses without fair distribution thresholds [19]. These findings are theoretically significant, yet they do not directly solve the farmer-mandi-day assignment problem that FPO managers face daily. The missing step is an algorithm that converts the idea of cooperation into a specific daily schedule that respects market absorption limits, transport realities, and urgency triggered by storage risk.

There is also a growing trend towards explainable AI and multilingual advisory interfaces for agricultural applications. SHAP and LIME have been integrated into crop yield prediction models achieving R^2 of 0.92 on Indian rice data [20], while counterfactual analysis has been applied to demonstrate actionable what-if scenarios for farmers [21]. In the chatbot domain, Farmer.Chat has achieved production-scale deployment using GPT-4 with Retrieval-Augmented Generation (RAG) across four countries, serving 15,000+ users via WhatsApp and Telegram [22], and AgroBuddy has demonstrated fine-tuned LLaMA-3 with F1 scores of 93% in Hindi and Marathi [23]. Region-aware retrieval techniques have addressed the contextual hallucination problem, where advice valid for one region may be harmful in another, achieving +0.12 F1 improvement over GPT-4-Turbo [24]. Efforts are also underway to leverage historical Kisan Call Center helpline records for automated farmer advisory, achieving 98.5% accuracy across 755 queries covering 151 crops [25].

While this technological progress across individual domains has led to impressive results, with various forecasting models reporting accuracy rates exceeding 92% [2], [8], [9], IoT systems achieving real-time monitoring at under ₹1,500 per node [11], [12], and chatbots successfully serving thousands of farmers [22], [23], a critical gap persists in integrating these components into a unified decision-support platform for cooperative selling. The existing literature provides surveys and individual solutions but fails to address the fundamental interdependence between where to sell, when to sell, and how much to sell — decisions that are currently managed

manually and in complete isolation within FPOs. Furthermore, a recent systematic review found that only 30% of India's 44,000+ FPOs are operating successfully, with digital platform adoption identified as a critical unmet need [26].

This paper aims to address this knowledge gap by providing a critical review of the current state of the art across six interconnected research streams and presenting a structured analysis that evaluates each stream's maturity against the requirements of a deployable cooperative selling platform. It highlights the specific limitations that prevent existing isolated systems from being combined into an integrated solution and offers concrete suggestions for future research, advocating for a shift from fragmented, component-level studies to unified, FPO-facing decision-support systems that treat forecasting, monitoring, coordination, explainability, and delivery as a single connected pipeline.

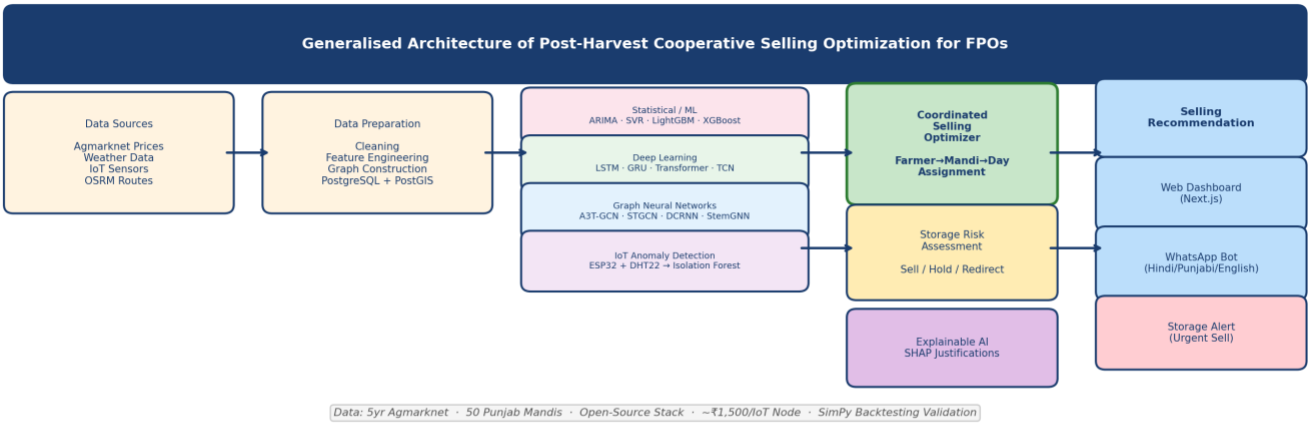

Fig. 1. Generalised architecture of post-harvest cooperative selling optimisation systems for FPOs.

Fig. 1 illustrates the generalised architecture of a cooperative selling optimisation system for FPOs, from data acquisition to recommendation delivery. Market prices and arrivals from Agmarknet, weather variables, IoT sensor readings, and transport route data are preprocessed and fed into forecasting models. These range from classical statistical and machine learning approaches (ARIMA, SVR, XGBoost) to deep learning methods (LSTM, GRU, Transformer, TCN) and graph neural network architectures (A3T-GCN, STGCN, DCRNN, StemGNN) that capture spatial price dependencies across mandi networks. In parallel, IoT data from ESP32-based storage sensors is processed through anomaly detection algorithms (Isolation Forest) to assess storage risk. The coordinated selling optimiser combines price forecasts with storage urgency constraints to generate farmer-mandi-day assignments that prevent arrival-induced price depression. Explainable AI modules (SHAP) provide justifications for each recommendation,

which are delivered through a web dashboard and a multilingual WhatsApp advisory interface supporting Hindi, Punjabi, and English.

2 Literature Review

The core research underpinning cooperative selling optimisation for FPOs spans six interconnected domains: agricultural price forecasting, spatiotemporal graph neural networks, IoT-based grain storage monitoring, post-harvest loss quantification and FPO dynamics, cooperative game theory for supply chains, and explainable AI with multilingual farmer advisory. Research in this field has evolved from isolated single-domain studies to increasingly integrated approaches, yet no existing system combines all six streams into a unified platform. The following subsections synthesise and critically analyse the relevant literature across these domains.

2.1 Agricultural Price Forecasting

The transition from statistical to deep learning models for agricultural price forecasting is now decisively established. Manogna, Dharmaji and Sarang (2025) conducted the largest Indian benchmarking study to date, evaluating eight models across 23 commodities and 165 AGMARKNET markets. GRU achieved a 76% RMSE reduction over ARIMA for onion prices (369.54 vs 1564.62), with MAPE ranging from 10–15% for volatile commodities [2]. This result confirmed that deep recurrent models consistently outperform statistical baselines on non-linear agricultural price dynamics.

Expanding on this understanding, Nayak et al. (2024) pioneered TransformerX and NBEATSX architectures incorporating weather variables from NASA POWER for Indian Tomato-Onion-Potato forecasting, achieving RMSE of 110.33 and demonstrating that weather-augmented deep learning models are robustly superior to univariate approaches [6]. Hybrid architectures have further improved accuracy. Zhao et al. (2025) evaluated a TCN-XGBoost model on 65,750 data points, achieving RMSE of 0.26 and MAPE of 5.3%, outperforming both standalone ARIMA (0.36) and LSTM (0.34) [7]. A significant milestone was achieved by Shaheen et al. (2025), who conducted the first known real-time deployment of deep learning for Indian fruit price prediction during the 2025 cherry season, achieving greater than 92% accuracy across five wholesale markets with Diebold–Mariano tests confirming statistical superiority [8].

More recently, Chaitra and Meena (2026) introduced a VAR-LSTM hybrid incorporating fuel prices as a covariate for Ragi price prediction, validating that transport-related variables significantly influence mandi-level pricing [9]. A 2026 MDPI study further demonstrated that integrating logistics costs, demand-supply ratios, and environmental factors yields superior prediction accuracy for Indian food crops using Gradient Boosting [10]. Despite these advances, all reviewed forecasting studies treat each market as an independent time series, ignoring the spatial price spillover between geographically proximate mandis. In Punjab, where trader movement and logistics create strong inter-market price correlations, this assumption is highly

restrictive. This gap directly motivates the adoption of graph neural networks for spatially aware price forecasting.

2.2 Graph Neural Networks for Spatiotemporal Forecasting

Özden and Bulut (2024) presented the first known application of GNNs to agricultural price forecasting, using StemGNN on Turkish wholesale vegetable prices (1,791 days, 2018–2022) and achieving MAE of 1.37 and RMSE of 1.94 while outperforming CNN, LSTM, and Random Forest [3]. The GNN captured inter-commodity price correlations analogous to spatial price spillover, validating the theoretical premise behind spatially aware agricultural forecasting. Building on this, Min et al. (2025) compared StemGNN, T-GCN, and Stacked LSTM on South Korean commodity prices with weather variables, confirming that GNN-based models outperform RNN-based models for multivariate agricultural price prediction. Notably, StemGNN's self-attention adjacency weights provided interpretability about which weather variables most strongly influence specific commodity prices [4].

In contrast to the agricultural applications, Mohammadzadeh et al. (2025) adapted A3T-GCN for air quality forecasting using a cosine similarity-based adjacency matrix instead of geographic proximity, reducing RMSE by 11% and MAE by 13% over standalone GRU [5]. This adjacency construction method is directly applicable to mandi networks, where markets may be connected by behavioural price similarity rather than physical distance. The critical research gap is clear: no study has applied A3T-GCN or any temporal GNN to Indian mandi networks. The two existing agricultural GNN papers target Turkish and Korean markets respectively, making the application to AGMARKNET mandi networks genuinely novel.

2.3 IoT-Based Grain Storage Monitoring

IoT grain monitoring using ESP32-class controllers has reached technical maturity. Alyammahi et al. (2025) demonstrated a fully integrated single ESP32-board system with DHT11, MQ135, and load cell sensors in a custom 3D-CAD polycarbonate enclosure, achieving ±2% sensor accuracy with Blynk IoT platform integration [11]. Sivathanu et al. (2023) addressed fatal CO_2 incidents in Indian silos by deploying ESP-WROOM32 with a three-tier alert system using Google Apps Script on free-tier Google Cloud [12]. These foundational systems established the viability of low-cost IoT monitoring but relied on simple threshold-based alerting.

To reduce the dependence on cloud connectivity, Karthiga and Thenmozhi (2024) deployed TinyML models directly on ESP32 for on-device classification of silo conditions, eliminating cloud dependency and enabling automated actuator responses even during connectivity outages [13]. For distributed facilities, a 2025 East African study deployed LoRaWAN-based sensors achieving 94% anomaly detection accuracy over 2,160 records, demonstrating superior communication range versus Wi-Fi [14]. A 2025 Foods (MDPI) study introduced a 3D DenseNet-based deep learning framework for spatial temperature field prediction in granaries, capable of identifying localised hotspots invisible to sparse sensor grids [15]. Most recently, a March 2026 study from Karnataka introduced a portable Smart Grain Management System

designed for continuous 6–12 month monitoring using ESP32 with DHT22 and MQ135 sensors, specifically targeting smallholder farmers with a mobile-first design [16].

However, in all reviewed systems, sensor data terminates at detection and alerting. No existing IoT monitoring platform connects storage condition data to selling decision logic. The farmer is informed that a problem exists but receives no guidance on whether to sell immediately, sell at a different mandi, or continue storage. This gap between monitoring and actionable market intelligence is a central motivation for integrated platforms.

2.4 Post-Harvest Losses and FPO Dynamics

Gulati, Das and Winter-Nelson (2024) conducted the first study estimating both quantity and quality losses for Indian crops, surveying 1,200 farmers across Punjab, Madhya Pradesh, and Bihar. Soybean exhibited the highest total loss at 15.34%, and critically, each kilometre increase in distance from the nearest mandi increases paddy loss by 1.4% [1]. This finding directly motivates transport-cost-aware mandi selection in cooperative selling platforms. Despite India forming 44,000+ FPOs, Patil et al. (2025) found that only 30% operate successfully, with 50% remaining in initial mobilisation phases. The study explicitly identifies digital platform adoption as the most critical unmet need for FPO sustainability [26]. These findings collectively establish the urgent institutional and economic justification for technology-enabled cooperative selling tools.

2.5 Cooperative Game Theory for Agricultural Coordination

Bekolli, Guardiola and Meca (2025) introduced a new class of totally balanced cooperative games for agricultural processing, proving that all agents are incentivised to cooperate through a non-empty core for all subgames [17]. This directly validates the premise that coordinated selling is individually rational for FPO members. Expanding on fair allocation, Chen et al. (2025) extended the Shapley value with TOPSIS-based corrective factors for risk-bearing, cooperation level, capital investment, and information access, demonstrating improved coalition stability in agricultural supply chains [18]. Shi and Cheng (2025) found that integrated logistics cooperation yields the highest profits but cooperation collapses without maintaining fair distribution thresholds [19].

These findings collectively inform the design of cooperative selling optimisers, which must ensure Gini fairness to prevent cooperative collapse. However, no study directly solves the farmer-mandi-day assignment problem that FPO managers face daily. The gap between game-theoretic incentive analysis and operational scheduling remains unfilled.

2.6 Explainable AI and Multilingual Farmer Advisory

SHAP and LIME have become standard tools for agricultural model transparency. Mohan et al. (2024) achieved R^2 of 0.92 using SHAP-integrated models for Indian rice yield prediction, revealing temperature as the most critical feature at 30% contribution [20]. A 2025 MethodsX study introduced counterfactual analysis alongside SHAP and LIME, demonstrating that changing specific features could shift predicted yield from 3.5 to 4.2 tons/ha — a practical

decision-support capability [21]. These studies confirm that SHAP-based justifications are technically feasible and increasingly expected by agricultural stakeholders. However, no study has applied XAI to price forecasting or selling recommendations, representing an application novelty.

In the chatbot domain, Farmer.Chat (Singh et al., 2024) represents the most mature deployment, using GPT-4 with RAG and 7,000+ farmer training videos to serve 15,000+ users across four countries via WhatsApp, with greater than 75% success rate [22]. AgroBuddy (Sheikh et al., 2025) fine-tuned LLaMA-3 70B with Sentence-BERT and Pinecone, achieving F1 of 93.0% in Hindi and Marathi with 1.8-second response time [23]. Fanuel et al. (2024) addressed contextual hallucination through geospatial metadata injection, achieving +0.12 F1 improvement over GPT-4-Turbo by ensuring region-specific advice is surfaced correctly [24]. Godara et al. (2024) leveraged 8 years of Kisan Call Center records to build AgriResponse with 98.5% accuracy across 755 queries covering 151 crops [25]. These advances provide a clear technical pathway for making selling recommendations transparent and accessible to Hindi/Punjabi-speaking farmers via WhatsApp.

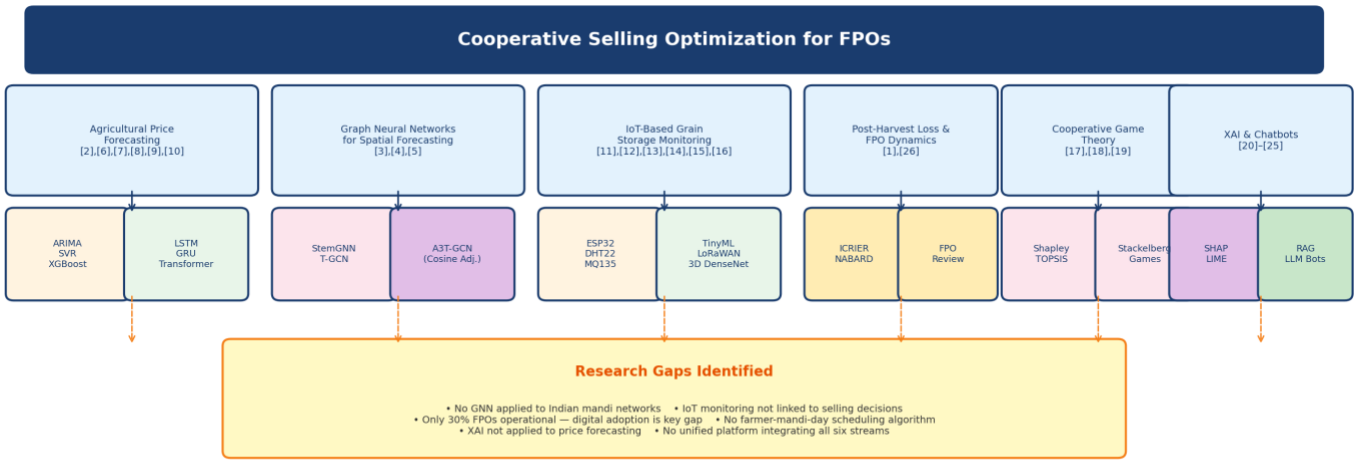

Fig. 2. Classification of research streams in cooperative selling optimisation for FPOs.

The studies reviewed show significant advancement across all six research streams, yet reveal a persistent fragmentation. Fig. 2 illustrates the major methodological categorisation and the research gaps identified at the intersection of these streams. Agricultural price forecasting has matured from ARIMA baselines to GRU/Transformer architectures achieving 76–92% improvement, but remains spatially unaware. GNN models have been validated for agricultural pricing in Turkish and Korean markets, but never on Indian mandi networks. IoT monitoring is technically mature but terminates at alerting without connecting to selling logic. Game theory proves cooperation is rational but does not produce scheduling algorithms. XAI and multilingual chatbots provide the delivery layer but lack integration with market intelligence. The synthesis of

these six mature but isolated streams into a single platform represents the core contribution targeted by this review.

Table 1. Summary of Literature Review Studies

Cite	Year	Problem Statement	Proposed Method	Results + Scores	Limitations
[1]	2024	Distance-to-mandi effects on post-harvest losses undocumented.	ICRIER farmer survey + regression analysis	Soybean 15.34% loss; +1km mandi distance = +1.4% paddy loss.	3 states only; policy-level; no technology solution proposed.
[2]	2025	Indian commodity price forecasting lacks large-scale DL benchmarking.	8 models (ARIMA→GRU) on 23 commodities, 165 mandis.	GRU RMSE 369.54 vs ARIMA 1564.62 (76% reduction).	Univariate only; no spatial modeling; no XAI.
[3]	2024	Inter-commodity price correlations ignored in forecasting models.	StemGNN (Graph Fourier + DFT) on Turkish vegetable prices.	MAE 1.37, RMSE 1.94; outperformed CNN, LSTM, RF.	Turkish market only; static graph; first agri-GNN.
[4]	2025	RNN vs GNN comparison for agricultural prices missing.	StemGNN, T-GCN, Stacked LSTM with weather variables.	GNN outperformed RNN; adjacency weights interpretable.	Korean market; limited commodities.
[5]	2025	A3T-GCN validated only in traffic — needs non-traffic domain proof.	A3T-GCN with cosine similarity adjacency for PM2.5.	RMSE −11%, MAE −13% over GRU; cosine adj. superior.	Air quality domain; 35 nodes; static adjacency.
[6]	2024	Indian vegetable forecasting ignores weather exogenous variables.	TransformerX + NBEATSX with NASA POWER weather.	RMSE 110.33, sMAPE 22.14; outperformed ARIMAX, XGBoost.	Only 2 exogenous vars; TOP crops only.
[7]	2025	Single models miss both temporal and nonlinear patterns.	TCN-XGBoost hybrid on 65,750 global data points.	RMSE 0.26, MAPE 5.3%; best among all models.	Global data only; no Indian mandi testing.
[8]	2025	No real-time DL deployment for Indian fruit markets exists.	LSTM live web system during 2025 cherry season.	Accuracy >92%; Diebold–Mariano confirmed superiority.	Cherry-specific; seasonal only.
[9]	2026	Fuel/transport costs ignored in Indian crop price models.	VAR-LSTM hybrid with fuel prices as covariate.	Outperformed standalone LSTM, GRU, CNN on all metrics.	Single commodity (Ragi); historical only.
[10]	2026	Multi-factor integration (logistics+economic+env) missing for India.	ML framework with transport cost + demand-supply ratio.	Gradient Boosting best; dim. reduction improved efficiency.	Limited dataset; no DL or spatial models.
[11]	2025	Grain storage lacks affordable single-board IoT monitoring.	ESP32 + DHT11 + MQ135 + load cell; Blynk IoT.	±2% accuracy; alerts at >27°C / >70% RH.	No ML; Wi-Fi dependent; 3-day battery.
[12]	2023	Fatal CO_2 incidents in Indian silos need automated response.	ESP-WROOM32 + Google Apps Script; 3-tier alerts.	Auto email + servo motor activation; free-tier GCP.	No ML; single silo; no market integration.
[13]	2024	Cloud-dependent IoT fails in remote	TinyML on ESP32 for on-device silo	Offline inference; automated	No accuracy metrics; not

		connectivity-limited areas.	classification.	actuator responses.	linked to selling.
[14]	2025	Distributed storage needs long-range low-power monitoring.	LoRaWAN sensors + ML anomaly detection (Kenya).	94% anomaly detection; superior range vs Wi-Fi.	Maize-specific; Kenyan context.
[15]	2025	3D spatial temperature prediction missing in grain monitoring.	3D DenseNet + 3DCNN-LSTM for granary risk alert.	Accurate 3D prediction; automated spatial risk alerts.	Chinese data; dense sensors needed.
[16]	2026	Indian smallholders need portable grain monitoring systems.	Portable ESP32 + DHT22 + MQ135; mobile-first.	6–12 month monitoring; automated alerts.	Prototype only; no ML or market integration.
[17]	2025	No proof cooperation is always optimal for agri-cooperatives.	Totally balanced cooperative games (Albanian apples).	Non-empty core proved; year-round selling enabled.	Apple only; no scheduling; Albanian context.
[18]	2025	Shapley value ignores risk, cooperation level, information.	Modified Shapley + TOPSIS corrective factors.	Platform share shifted 15.7%→19.7%; stable coalition.	Simulation only; 3-player; Chinese context.
[19]	2025	Optimal logistics cooperation structures undefined.	Four Stackelberg game models compared.	Integrated cooperation highest profit; collapse threshold found.	E-commerce focus; no mandi scheduling.
[20]	2024	Agri-ML models are black boxes — no farmer trust.	SHAP + LIME on DT, RF, LightGBM for rice yield.	R^2 0.92; temperature 30%, rainfall 45% contribution.	Rice yield only; not price forecasting.
[21]	2025	Ensemble models sacrifice interpretability for accuracy.	RF+LSTM+XGBoost + SHAP + LIME + counterfactual.	Counterfactual: yield shifts 3.5→4.2 tons/ha.	Yield only; not tested on price data.
[22]	2024	Smallholder farmers lack multilingual advisory at scale.	GPT-4 + RAG; 7,000+ videos; WhatsApp/Telegram.	15,000+ users; 300,000+ queries; >75% success.	English intermediary; no market intelligence.
[23]	2025	Indian farmers need local-language domain-specific advisory.	LLaMA-3 70B + RAG + Sentence-BERT + Pinecone.	F1 93.0%; BLEU 0.82; 1.8s response time.	Hindi/Marathi only; no Punjabi; GPU required.
[24]	2024	LLM chatbots hallucinate across agro-climatic regions.	Geospatial metadata injection + region-prioritised RAG.	F1 +0.12 over GPT-4-Turbo; hallucinations −10–20%.	160 questions only; English evaluation.
[25]	2024	8 years of Kisan Call Center data underutilised.	Three Response Retrieval Models with spatial params.	98.5% accuracy; 755 queries; 151 crops; open-source.	English-only; crop disease only.
[26]	2025	FPO success rates and digital needs poorly documented.	Systematic review + bibliometric analysis of 64 studies.	30% FPOs successful; digital adoption = key gap.	Backward-looking; no tech solutions proposed.

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